Pragnesh Dubey — Technical Product Manager Assessment (FactWise)

Product: QuarantineOS — a command center for quarantine facility operations (nurse temperature rounds → doctor rounds → discharge → outcome quality).

Q1 — User types and responsibilities

Capability	Nurse	Doctor	Administrator	Facility Manager
Dashboard / bed map / next-best-actions	View	View	View	View
Patient registry (search, history, timeline)	View	View	View + admit	View (read-only)
Record daily temperature (°C)	**Yes**	No	No	No
Correct a temperature (reason required, audited)	**Yes**	No	No	No
Complete doctor visit + clinical notes	No	**Yes**	No	No
Emergency override of visit block (reason, audited)	No	**Yes**	No	No
Record discharge recommendation (clinical clearance)	No	**Yes**	No	No
Confirm actual discharge / free bed	No	No	**Yes**	No
Admit patient (if bed free)	No	No	**Yes**	No
Record death outcome (reason, audited)	No	No	**Yes**	No
Outcomes & Quality, audit log	View	View	View	View
Settings: fever threshold, capacity	No	No	Yes	**Yes**

Essential screens per role. Nurse: Temperature Rounds (pending/completed list, record form). Doctor: Doctor Rounds (queue, latest temperature, streak, visit form, recommend discharge). Administrator: Discharge Management, Patient Registry (admission), Outcomes. Manager: Dashboard, Outcomes & Quality, Audit Log, Settings. Rationale: each person sees one clear queue, so no one has to guess what has already been done — this directly removes the duplicate measurements, missed measurements, premature doctor visits and late discharge hand-offs described by the head doctor.

Q2 — Functional prototype

Live prototype: https://claude.ai/artifact/E3ZR12adFN6jbjbWq8C2pf
Hosted as a Claude artifact; it is private to the author until shared via the artifact share control. If it does not open for you, I will share the link or the single-file source (quarantineos.html).

What it is. A single-page app with 8 screens: Operations Dashboard, Patient Registry, Temperature Rounds, Doctor Rounds, Discharge Management, Outcomes & Quality, Activity & Audit Log, Settings/Demo Guide. A top-bar **role switcher** (Nurse / Doctor / Administrator / Facility Manager) lets the evaluator try every workflow without accounts. Permissions are enforced in the logic layer, not only by hiding buttons.

Core workflow to try. 1) Nurse → Temperature Rounds → record 37.1 for P-002 (task flips to Recorded; dashboard updates). 2) Doctor → Doctor Rounds → P-002 shows the new temperature; complete visit with notes. P-009 shows “Why blocked?” because no temperature exists today. 3) Doctor → Recommend discharge for P-007 (3 fever-free days). 4) Administrator → Discharge Management → Confirm discharge; occupancy drops 74→73 and an audit entry appears. 5) Registry → admit a new patient into the freed bed. 6) Outcomes & Quality → 20 sample outcomes (4 deaths) show 20% mortality vs 15% benchmark with a warning, 80% success vs 85%.

Differentiators. Next-best-action queue ordered by what frees beds first; 3-dot fever-free progress indicator; “Why blocked?” explanations; bottleneck indicator (missing temperatures vs waiting visits); patient-flow timeline; 74-bed map.

Limitations (honest). Demo data is fictional and stored only in the browser (localStorage) — not secure, not multi-user, not production persistence. Role switching is not authentication. The fever threshold (default 38.0 °C) is a **demonstration setting that needs clinical confirmation**. Not deployed on Vercel (no deploy access in my build environment). Tested with automated logic tests and a simulated-browser smoke run; I did not capture screenshots, so none are included.

Q3 — Logical constraints for developers

Area	Rule (enforced in prototype unless marked “Prod”)
Capacity	Admitted patients never exceed capacity (74). Admission with a full facility is rejected with a message to discharge first. Capacity cannot be set below current occupancy. Bed auto-assigned to first free bed. Prod: transfer/bed-cleaning states.
Patient data	Name and age (0–120) required; duplicate patient IDs rejected; status transitions only admitted→discharged or admitted→deceased; closed patients cannot receive tasks.
Temperature	Celsius, valid range 34.0–43.0 °C, one reading per patient per local calendar day. A second entry is blocked unless it is a correction with a reason (≥5 chars); corrections keep before/after values and are audited. Fever flag when ≥ configurable threshold. Warning banner lists patients with no reading today; “overdue” shown after 12:00 (demo rule). Prod: device integration, multiple readings with a designated official one, time-zone handling.
Doctor visit	Routine visit cannot be completed unless today’s temperature exists (shows “Why blocked?”). Notes required. One visit per day. Doctor-only emergency override needs a documented reason (≥10 chars) and is flagged in the audit log. The system displays data and never prescribes treatment.
Fever-free streak	Count consecutive complete calendar days ending today (if measured) or yesterday, each with a valid reading below threshold. A missing day breaks the streak and never counts as fever-free; a fever resets it to 0. Recomputed on every read, so entry, correction or threshold change updates it instantly. Assumption: a day with a recorded below-threshold reading counts as a “day”.
Discharge	Recommendation allowed only when streak ≥ 3. Administrator discharge requires streak ≥ 3 **and** an active clinical clearance. If a fever is recorded after clearance, clearance shows “withdrawn” and discharge is blocked. Nothing is auto-discharged. Confirmation dialog, bed freed, outcome “cured” recorded, audit entry written.
Outcomes	Resolved = cured + deceased; mortality = deceased ÷ resolved; success = cured ÷ resolved; active patients excluded. No resolved outcomes → “Insufficient outcome data”. Red warning when mortality > 15%; success below 85% highlighted. Descriptive only — no claim of statistical significance or causality. Death recording: Administrator only, reason required, confirmation.
Duplicates / metrics	Workflow counts use distinct patients per day, so repeat actions cannot inflate completion. All dashboard numbers derive from the same state.
Permissions	Every action checks the role in logic; unauthorized attempts return an error and change nothing. Prod: real authentication, per-nurse/doctor identity, ward-level assignment, session timeout.
Audit	Logged with timestamp, role, patient, action, reason: admission, temperature, correction, visit, override, recommendation, discharge, outcome, settings changes. Prod: append-only, tamper-evident, user-level identity, retention policy.

Automated checks run: 17 business-rule tests (capacity, invalid temperature, duplicates/corrections, missing days, fever reset, 3-day eligibility, visit block + override, role rejection, discharge gating, occupancy, mortality/success, empty denominator) — all passed; plus a simulated-browser run of the nurse→doctor→admin journey with no runtime errors.

Q4 — Sprint prioritization

Budget: velocity 25 – 3×5 committed = **10 points**. **Values:** High = 6, Medium = 3, Low = 1 (High = 2× Medium, Medium = 3× Low). Additive value model; effort budget 10.

Feature	Priority	Effort	Value	Value/effort
A	High	1	6	6.0
B	High	3	6	2.0
C	High	5	6	1.2
D	Medium	2	3	1.5
E	Medium	3	3	1.0
F	Medium	1	3	3.0
G	Low	0.4	1	2.5
H	Low	0.5	1	2.0
I	Low	0.3	1	3.3
J	Low	0.1	1	10.0

Method. This is a 0/1 knapsack. Total effort of all ten features is 16.3, over the 10 budget, so I enumerated all 1,024 subsets and kept those with effort ≤ 10 (exhaustive check, not a greedy heuristic).

Result — a correction. The maximum value is **22**, not 21. The combination A + B + C + F (effort 10, value 21) is feasible and good but **not optimal**. Optimal combinations:

Combination	Effort	Value
A, B, D, F, G, H, I, J (recommended)	8.3	22
A, B, E, F, G, H, I, J	9.3	22

Why: the four Low features together cost only 1.3 points for 4 value (more than one Medium for 1 point). Dropping the 5-point C (value 6) frees budget for D (3) plus all four lows (4) — a net gain of +1. Several other 21-value sets also exist (e.g. A+B+C+F; A+B+C+G+H+J), but all are dominated.

Recommendation. Take A, B, D, F, G, H, I, J: value 22 at 8.3 points, leaving 1.7 points of buffer (good for estimation risk, which matters when velocity is only an estimate). Trade-off: C, a High-priority feature, is deferred.

Math vs reality. The model assumes value is additive, estimates are exact, and features are independent. In practice I would first check dependencies (does A, B or D need C?), risk and uncertainty, and whether C unblocks a business-critical or safety-critical capability (e.g. if C were the discharge-clearance workflow, I would keep C and drop lower-value items). Eight features in one sprint also adds context-switching and testing overhead, so I would validate the plan with the team before committing.